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Mechatronics Systems Design

Smart Trash Bin – V Model Analysis

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Client requirements

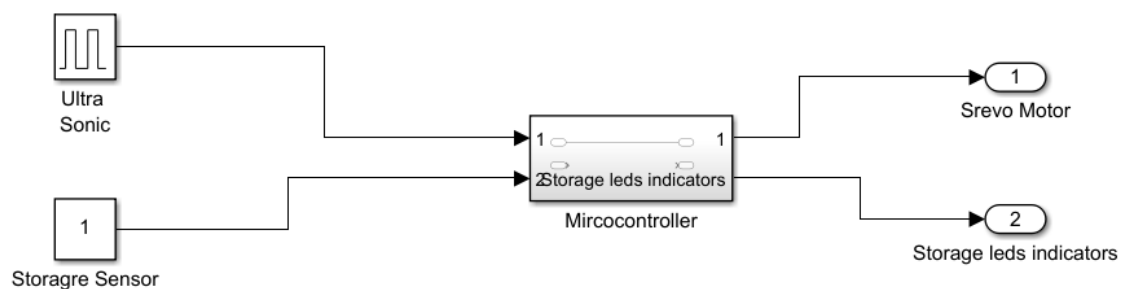
1. The trash bin should **open automatically when a person's hand approaches the lid.**
2. The lid should **close automatically after a few seconds.**
3. The system should **detect when the bin is full.**

System Requirements Development and analysis

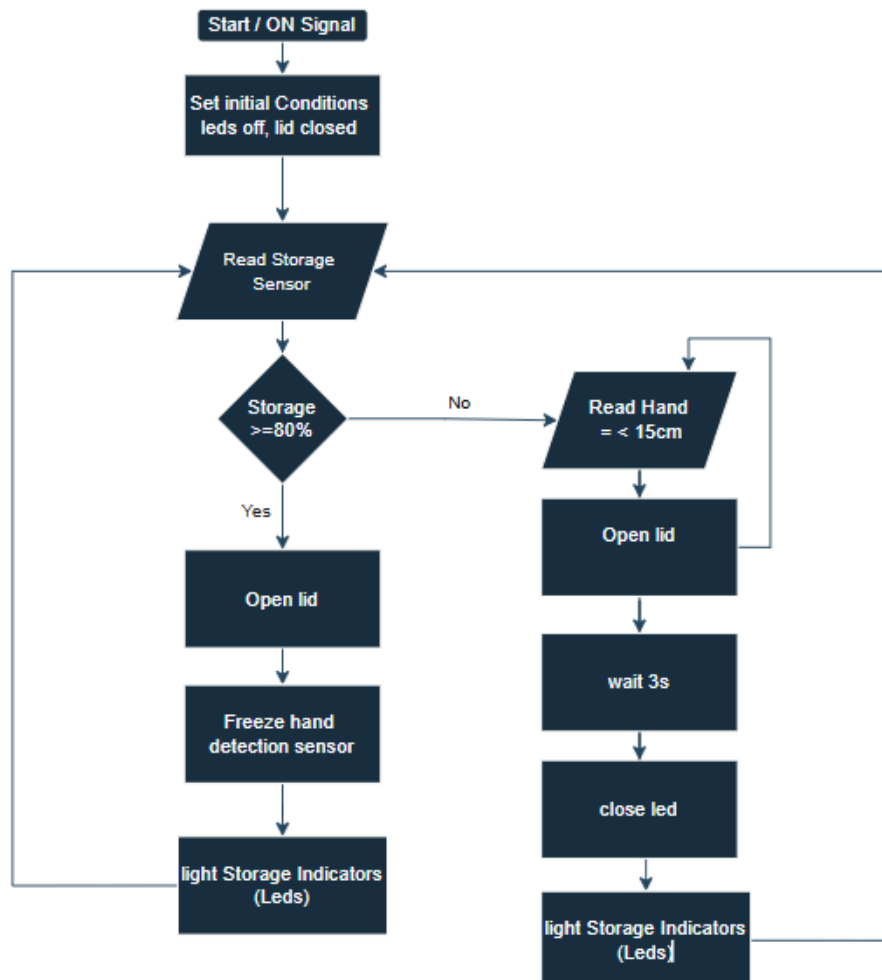
Requirements	Applicability	Comments
Object detection	Applicable	We must be careful about false triggers due to noise
Opens Automatically (Actuation)	Applicable	We must be careful about false triggers due to noise
Closes Automatically (Actuation)	Applicable	Time to close the lid must be tested for convenience
Storage detection	Applicable	Need to confirm if only full detection required or full level detection
Off / On Switch	Applicable	Extra but recommended
Hand detection sensor goes in sleep mode When the storage is full to save power.	Applicable	Extra but recommended

System Architecture

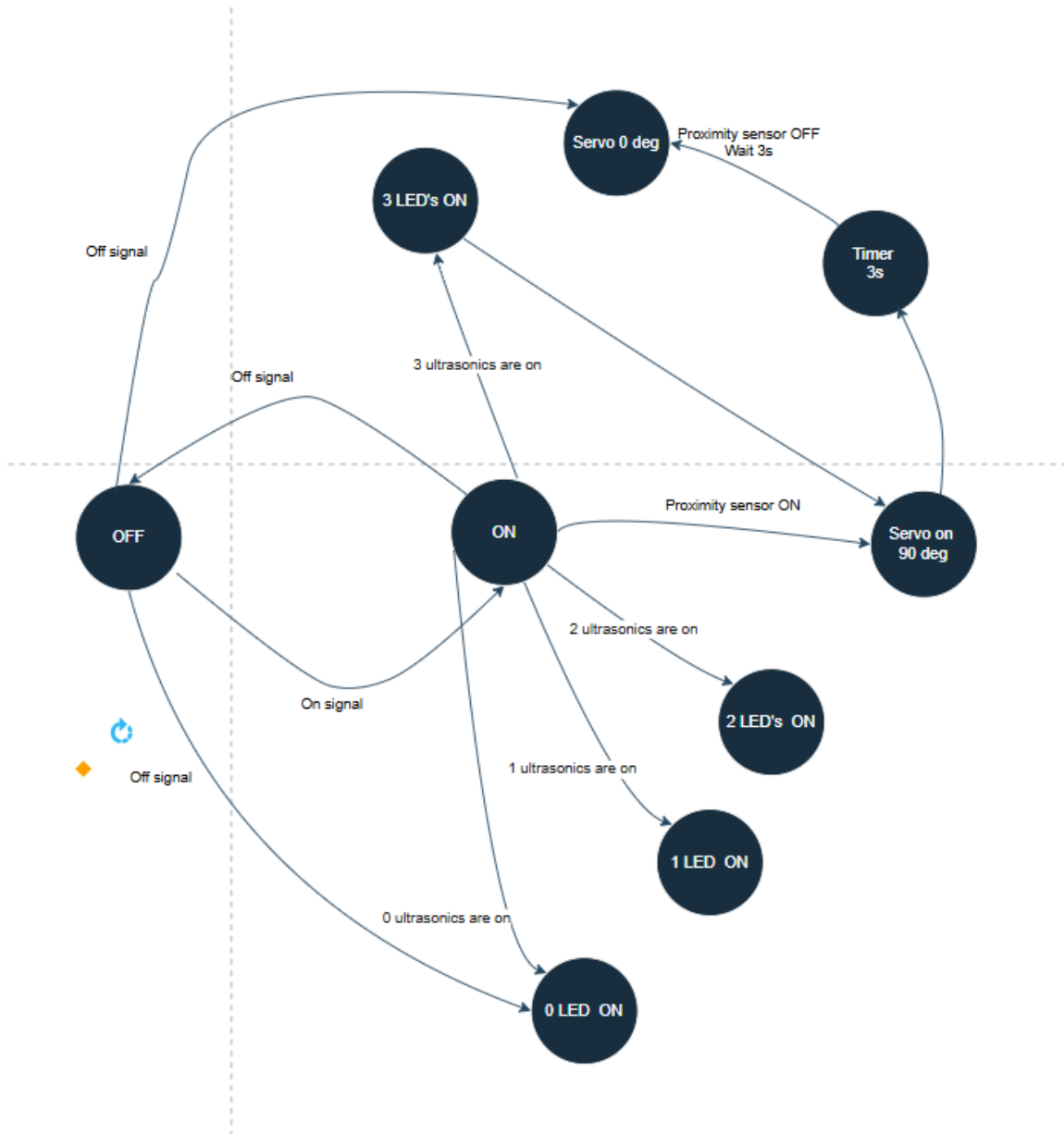
Block Diagram



Flow chart



State Machine Diagram



Subsystems & hardware.

The architecture is divided into four modular subsystems:

1. **Sensors Subsystem:** Responsible for gathering data detecting hands and measuring trash volume.

2. **Actuation Subsystem:** Responsible for the physical mechanical movement of the lid, which is servo motor.
3. **Control Subsystem:** Microcontroller unit and control switches.
4. **Power Subsystem:** 12v replaceable batteries, DC – DC buck convert and Decoupling Capacitors.

The hardware and software components are listed as follows:

Components	Description
Microcontroller unit (MCU)	Arduino Nano, serves as the central processing unit to read sensor inputs and execute control logic for the system.
MCU software	Arduino IDE, used for writing, compiling, and uploading the C++ control firmware to the MCU.
Storage sensors	3 Waterproof Ultrasonic (JSN-SR04T) sensors will be placed along the trash bin; each one will indicate a certain level.
3 LED's	3 red light emitting diodes that will be an indicator for each storage sensor level.
IR Proximity Sensor	A (TCRT5000) sensor for hand detection.
Motor	Servo Motor (MG996R), it operates on 5V to 6V, has robust metal gears, and provides about 10-11 kg-cm of torque.
Control rocker Switch	A standard SPST (Single-Pole Single-Throw) rocker switch rated for at least 3 Amps to cut power directly from the 12V battery line.
Battery	A 3S 18650 Lithium-Ion Battery Pack (11.1V - 12.6V), the cells can be easily swapped and charged in a standard 18650 charger.
DC-DC Buck Converter	LM2596 Step-Down Module , a highly reliable and cheap switching regulator. You will wire the 12V battery to the input and turn the onboard potentiometer until the output reads exactly 5V.
Decoupling Capacitor	1000μF Electrolytic Capacitor (16V or 25V rated), placed across the 5V and GND lines as close to the servo as possible to absorb the inductive kickback and current spikes when the lid opens.

Test Cases

Hardware Subsystems Calibration and Tests

Test Cases ID	Test Scenario Test Steps Expected Result	Test Scenario Test Steps Expected Result	Test Scenario Test Steps Expected Result



HW-CAL-01	Ultrasonic Sensors (JSN-SR04T) Calibration	Place a flat, rigid object at exact known depths (e.g., 10cm, 30cm, 50cm) inside the bin. Monitor the raw distance output via the Arduino Serial Monitor.	Sensor readings match physical measurements within ± 1 cm.
HW-CAL-02	IR Proximity Sensor (TCRT5000) Calibration	Place a hand at the desired trigger distance (e.g., 15 cm), create logic for false triggering (average filter). Adjust the onboard potentiometer on the IR module until the indicator LED triggers.	Sensor reliably switches state (HIGH/LOW) exactly at the threshold without false triggers from ambient room lighting.
HW-PWR-01	DC-DC Buck Converter Output Tuning	Connect the 12V 3S battery to the LM2596 input. Connect a multimeter to the output and adjust the screw terminal.	Output voltage is stable at exactly 5.0V to 5.5V.

Software (SW) Test Cases

Test Cases ID	Test Scenario Test Steps Expected Result	Test Scenario Test Steps Expected Result	Test Scenario Test Steps Expected Result
SW-TC01	Volume Logic	Filling the trash bin, and monitoring ultrasonic readings, and correct leds outputs	The system correctly activates 1, 2, or 3 LEDs corresponding to the simulated thresholds.
SW-TC02	Actuation State Machine	Trigger the openLid() function, wait for	The PWM signal moves the servo to



		the programmed delay (e.g., 5 seconds), then trigger the closeLid() function.	the open position, holds, and returns to the closed position.
SW-TC03	Sensor Averaging Algorithm	Feed fluctuating dummy distance values into the volume calculation function to simulate a moving or uneven trash pile.	The software successfully applies a rolling average, to prevent false triggering.

Systems Test Cases

Test Cases ID	Test Scenario Test Steps Expected Result	Test Scenario Test Steps Expected Result	Test Scenario Test Steps Expected Result
SYS-TC01	Full Operation Test	Approach the bin, wait for the lid to open, drop an item inside, step away, and observe the system.	Lid opens smoothly, closes after a set delay, and the LED volume indicators update to reflect the newly added item.
SYS-TC02	Triggering Test 1	Keep a hand continuously in front of the IR sensor while the lid is open.	The lid remains open and the closing timer constantly resets; it does not attempt to close on the user's hand
SYS-TC03	Triggering Test 2	Try to trigger the IR fast to imitate insects movements, or sudden unintentional movements.	The lid should not open due to very fast triggers.
SYS-TC04	"Bin Full" Action	Fill the bin until the 3rd LED (Full) is lit.	The lid opens to indicates that it is full, and all LEDs



		Attempt to trigger the IR sensor.	light to alert the user.
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